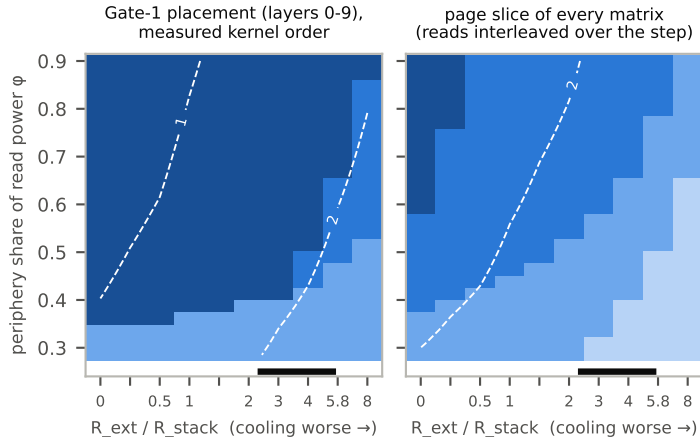


(a) cheapest model within $\pm 10\%$ of the macro-resolved allowable E/bit at 19 TB/s

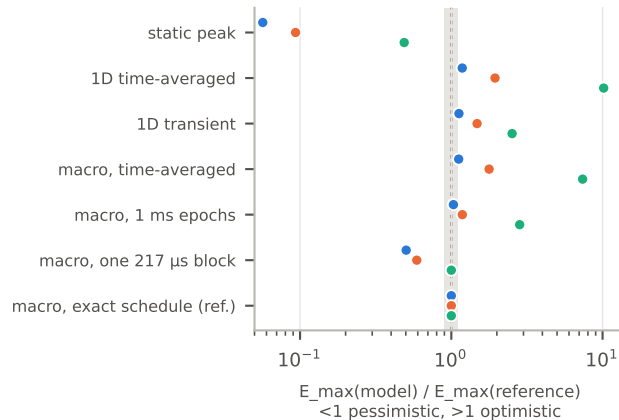
(b) boundary error by abstraction, $\phi = 0.9$

dashed: allowable E/bit [pJ/bit]; bar: B200-class liquid cooling;
0.217 pJ/bit passes everywhere on both maps



cheapest adequate model

- 1D time-averaged
- 1D transient
- macro-resolved, 1 ms epochs
- full spatiotemporal



representative points ($R_{\text{ext}}/R_{\text{stack}}$)

- interleaved reads, production film (5.8)
- Gate-1 placement, liquid cooling (3.0)
- one contiguous block, die-proximate cooling (0)